THE UNITED REPUBLIC OF TANZANIA
PRIME MINISTER'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
MPWAPWA DISTRICT COUNCIL

FORM TWO PRE MOCK II EXAMINATION
031 - PHYSICS

TIME: 2:00 HRS **MARCH 2026**

Instructions

1. This paper consists of sections A, B and C with a total of **ten (10)** questions.
2. Answer **all** the questions in each section.
3. All answers must be written in the spaces provided.
4. All writing must be in **blue** or **black** ink, **except** drawings which must be in pencil.
5. Communication devices and any unauthorized materials are **not** allowed in the assessment room.
6. Write your **NAME at** the top right corner of every page.
7. Where necessary the following constants may be used:

Acceleration due to gravity, $g = 10\ m/s^2$; $\pi = 3.14$; Density of water, $\rho_w = 1000\ kg/m^3$

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SECTION A (15 Marks)
Answer all questions in this section.

1. For each of the items (i) − (x), choose the correct answer among the given alternative and write its letter in the box provided.

i) When a capacitor is charged and disconnected from a battery what will happen to its charge and voltage if the distance between the plates is increase?
   A. The charge will remain constant but the voltage will increase
   B. Both charge and voltage will decrease
   C. The charge will decrease but the voltage will remain constant
   D. The charge will remain constant but the voltage will decrease.

ii) If you want to detect the presence of electric charge in an object, which electrical device would you use?
   A. Electrophorus C. Voltmeter
   B. Galvanometer D. Electroscope

iii) Imagine two groups of equal number of people are pulling a rope by their bare hands, each group exert on the rope?
   A. Tension force C. Frictional force
   B. Normal force D. Stretching force

iv. Why is it important to study physics in real life situations?
   A. It is useful for career development.
   B. It is used to clarify the existence of historical sites
   C. It explains physical properties of matter
   D. It is applied in the manufacturing and designing of the electronics devices
   E. It helps in experimental discoveries.

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A. 1, 2, 3 and 4
B. 2, 3, 4 and 5
C. 1, 3 4 and 5 D. 1, 2 4 and 5

v) A needle may float on a clean water but sinks when some detergent is added to water because the detergent
   A. Reduces the density of water
   B. Increase adhesive force between the needle and water molecules.
   C. Lowers the surface tension of water.
   D. Makes water surface slippery.

vi) A static electricity is the accumulation of electric charges
   A. In motion
   B. at rest
   C. in a circuit
   D. battery

vii) If you are caught outside during a severe thunderstorm, you should
   A. Take shelter under the nearest tree
   B. Stand under power lines
   C. Move to higher ground
   D. Hide in a ditch

viii) A mass of 0.2kg produces an extension of 8cm in a spring. The force required to produce an extension of 6 cm is
   A. 0.75 N
   B. 2.70 N
   C. 1.50 N
   D. 24.00 N

ix) Pressure in a liquid is independent of the
   A. Density of the liquid.
   B. Depth below the surface of the liquid.
   C. Pressure exerted on the surface of the liquid above.
   D. Cross sectional area and the shape of the vessel containing the liquid.

x) A student measured the diameter of a wire Using a screw gauge and he repeated the experiment four times and listed the measurements. The correct measurement was
   A. 5.3 cm C. 5.3200 cm
   B. 5.32 cm D. 5.320 cm

2. Match the meaning of the terminologies used in structure and properties of matter in **LIST A** With their corresponding terms in **LIST B** by writing the letter of the correct answer in the Table provided.

List A	List B
(i) Tendency of matter to be in state of random motion	A. Surface tension
(ii) Ability of liquids to rise or fall in a narrow tube	B. Cohesive
(iii) Attraction force between molecules of the same substance	C. Adhesive
(iv) Ability of the surface of a liquid to behave like a fully stretched elastic skin	D. Diffusion
(v) Movement of particles from high concentration to low concentration	E. Osmosis
	F. Capillarity
	G. Brownian motion

SECTION B (70 Marks)

Answer all questions in this section.

3. a) State any four applications of atmospheric pressure
b) Briefly explain why the woman wearing shoes with pointed heels is more likely to cause damage to the wooden floor than the elephant.
c) What will be the pressure due to a column of water of height 3m if the density of water is 1000 kg/m3.

4. a) A form two students performed an experiment on determination of the acceleration due to gravity by using an iron metal ball. With diagram briefly describe the energy changes when the bob of a Simple pendulum swings from one side to another.
b) (i) Which phenomenon is taking place when kerosene rises up a wick?

(ii) A force of 9.6N stretches a spring by 6cm while a force of 14.4N stretches it by 9cm. What force would be required to stretch the spring by 15cm

5. A rectangular block measures length 4.00cm, width 2.50cm and height 1.00cm.
a) Name the instrument used to measure the sides of the rectangular blocks
b) Calculate the volume of the rectangular block.
c) If the mass of rectangular block is 300g, find the density of the block.

6. a) What are the differences and similarities between the following pairs of terms?

(i) Speed and velocity

(ii) Distance and displacement

b) i) Sketch the diagram of a body which starts from rest and accelerates uniformly for some time to a constant velocity and then maintains this velocity for a certain period of time before decelerating uniformity to stop.

(ii) A car moving with a uniform velocity of 100 m/s is decelerated at 2.5 m/s to a stop. calculate, time taken for the car to stop and distance traveled by the car before it is brought to rest.

7. a) Describe what happens when an ebonite (or plastic) rod is rubbed with fur. what charge does the rod gain, and why?
b) Identify those which would become negatively charged for each of the following pairs of materials being rubbed together.

i) Glass and silk _____ ii) Fur and glass _____ iii) Comb and hair _____

iv) Fur and hard rubber _____

8. a) Explain how lighting is formed?
c) Describe how a lighting conductor protects a building.
d) Describe three safety measures during a lightning and thunderstorms

9. a) State the law of polarity
b) Sketch the lines of force between two bars of magnet pressed on horizontal table with:

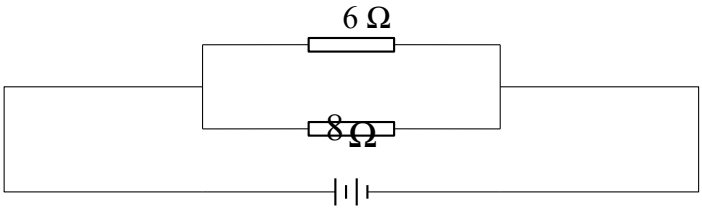
i) Their north poles facing each other

ii) One north pole facing the south pole of another

SECTION C (15 marks) Answer all questions in this section

10. a) State ohm's law

b) The circuit diagram below shows that the current through the 6 Ω resistor is 2.5 A



What is the magnitude of the voltage from the power supply?

(i) Determine the current through the 8 Ω resistor

c) The form two students are given a set of materials in physics Laboratory as following Rheostat, voltmeter, ammeter, fixed resistor, switch and a cell to perform experiment aimed to verify ohm's law. Draw a complete circuit diagram to show the arrangement such that the current can flow